

NEUROLENS: MULTIMODAL AI FRAMEWORK FOR EARLY PARKINSON'S DISEASE DETECTION

RP 25-26J-109

Project Proposal Report

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August 2025

EARLY DETECTION OF PARKINSON'S DISEASE VIA TOUCHSCREEN DRAWING ANALYSIS

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date

2025-08-31
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Abstract

Parkinson’s disease is a progressive brain disorder that affects both movement and visual–spatial abilities. This study explores an approach for early detection of these difficulties by analyzing spiral and wave drawings made on touchscreen devices. The goal is to develop a reliable and accessible diagnostic tool that can identify early warning signs before the well-known motor symptoms become apparent.

The method uses an AI-based framework that combines both image features and movement patterns, such as drawing speed, line smoothness, pressure changes, and overall complexity, from drawings produced by patients at different stages of the disease. Deep learning models, including Convolutional Neural Networks (CNNs) and Vision Transformers, are used to detect early motor problems, while repeated testing over time helps track disease progression.

To make the system trustworthy for clinical use, explainable AI techniques are applied to show how the model makes decisions. Unlike traditional assessments that focus only on visible motor decline, this approach captures the timing and dynamics of movement and can be used for real-time screening on mobile devices. Because it is low-cost, non-invasive, and scalable, the method has strong potential for early diagnosis, continuous monitoring, and improved care for patients with neurodegenerative diseases.

Keywords: Parkinson’s Disease, early detection, digital biomarkers, spiral and wave drawing analysis, touchscreen devices, deep learning, Explainable AI, mobile health, longitudinal monitoring

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
PD	Parkinson's Disease
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
Grad-CAM	Gradient-weighted Class Activation Mapping
SHAP	SHapley Additive exPlanations
UPDRS	Unified Parkinson's Disease Rating Scale
MDS-UPDRS	Movement Disorder Society – Unified Parkinson's Disease Rating Scale
TinyML	Tiny Machine Learning
SUS	System Usability Scale
HL7	Health Level Seven International (data exchange standard)
WBS	Work Breakdown Structure

1. INTRODUCTION

1.1. Background and Literature Survey

Parkinson's disease (PD) is the second most common neurodegenerative disease in the world, with about 10 million patients and increasing incidence with age exponentially associated [1]. It is pathologically defined by progressive dopaminergic neuronal loss in the substantia nigra, which culminates in its ultimate expression as the classical tetrad of motor symptomatology, namely tremor, rigidity, bradykinesia, and postural instability [2]. Interestingly, the insidious and creeping nature of PD is such that only after a large neuron loss will clinical manifestation of the motor signs be seen, and an estimated 50–70% of dopaminergic neurons have been lost before symptomatology can be noted [3].

Diagnosis increasingly relies on clinical observation and use of generic rating scales such as the Unified Parkinson's Disease Rating Scale (UPDRS) and Movement Disorder Society revision (MDS-UPDRS) [4]. While they have been widely used, they suffer from subjectivity, inter-rater reliability, and failure to detect subtle impairment in the prodromal period—i.e., when earliest treatment would be most valuable [5].

1.1.1 Digital health revolution in neurodegenerative disease assessment

The rapid expansion of digital health technologies has driven a paradigm shift towards the assessment of neurodegenerative diseases, with the provision of objective and quantifiable alternatives to observation-based methods [6]. Digital biomarkers—quantifiable physiological and behavioral data derived from digital devices have shown substantial potential in the early identification of Parkinson's disease (PD) for the applications of gait analysis, voice and speech assessment, and evaluation of fine motor control [7].

In this case, handwriting and drawing analysis are especially helpful as they are highly sensitive to fine motor impairment that is characteristic of PD. Drawing protocols call upon several neuromotor pathways to be activated, including the basal ganglia, cerebellum, and motor cortex—brain structures maximally impaired in the disease [8].

Spiral drawing has been specifically leading among such protocols as it provides the possibility of simultaneous recording of kinematic and spatial parameters, indicating motor impairment and disease severity.

1.1.2 Technological evolution and accessibility

The history of digital drawing analysis has traveled far, from being specialized digitizer tablets used only in the clinic to being widely available consumer-standard touchscreen devices [9]. Smartphones and tablets of today with high-definition displays and advanced sensors now enable sampling rates of 120–240 Hz sufficient to detect slight motor fluctuations but accessible and portable [10].

This technological innovation has enabled greater accessibility and scalability, leading the way to population-level screening and remote monitoring in real time. These characteristics are particularly valuable in the case of Parkinson's disease, wherein early diagnosis and precise monitoring of symptom onset are critical to successful clinical management and timely treatment [11].

1.1.3 Current approaches in digital drawing analysis

Recent studies extensively applied spiral drawing analysis as a diagnostic tool of PD with promising results based on ubiquitous statistical examination and deep learning methods. San Luciano et al. found that spirals in PD patients were stereotypically disorganized, slower, and had tighter curvature in comparison to controls, diagnostic utility sustained throughout the different stages of the illness [12].

Current machine learning methods have also played a part in driving this research forward. Saleh et al. created a CNN system for spiral image analysis based on handwriting with more than 90% accuracy in institutional data sets [13]. Alam et al. also proposed early PD diagnosis systems based on handwriting and drawing analysis, utilizing deep learning models with cosine annealing schedulers and transfer learning methods to achieve maximum efficiency [14].

1.1.4 Multimodal assessment approaches

Recent research results have employed multimodal methods with a range of drawing tasks to improve the classification performance. For instance, comparative experiments of wave and spiral drawing were performed by Bernardo et al. using DenseNet201 and VGG16 models with satisfactory performances in Parkinson's disease classification [15]. These deep learning models, the majority of which are, however, computationally very expensive and require a substantial amount of data to learn effectively, thereby constraining their application in resource-constrained clinical settings.

To address the limitations mentioned above, recent research has attempted to integrate kinematic features with spatial analysis. Parameters such as pen-pressure estimation and spiral speed of drawing have been put forward by Zham et al., enabling useful discrimination between PD stages [16]. The research emphasized the balance of temporal dynamics with non-temporal features in space and reasserted the importance of an integrated approach in offering adequate and clinically significant Parkinson's disease diagnosis.

1.1.5 Technological platform evolution

The technological environment of digital drawing analysis has significantly changed in the last twenty years. The first studies were based on high-end digitizing tablets with high sampling frequencies (>200 Hz) and pressure sensitivity, enabling accurate measurement of fine motor skills, but not easily accessible because of high prices and specific requirements for setting them up [17]. More recent studies have shifted directions toward consumer-grade smartphones and tablets, thus democratizing the use of digital biomarker collection without diminishing diagnostic effectiveness [18].

Burq et al. further demonstrated that virtual Parkinson's disease exams can be done using affordable consumer hardware to facilitate normal and regular remote monitoring of motor function [19]. The research pipeline of such an initiative not only opens the door for scalable digital health solutions but also necessitates validation of such an approach across various populations and technology platforms to introduce generalizability and clinical mastery.

1.1.6 Explainable AI in medical diagnostics

The use of explainable artificial intelligence (XAI) techniques in medical applications has become increasingly needed to facilitate clinical trust and adoption. One of the most popular techniques is SHAP (SHapley Additive exPlanations), a technique developed by Lundberg and Lee, that consolidates a framework for model prediction interpretation by quantifying the contribution of each feature to diagnostic outcomes [20]. To augment this, gradient-weighted class activation mapping (Grad-CAM) enables visualization of a convolutional neural network output by highlighting regions of input data that have the largest contribution to classification results [21]. These techniques, in conjunction, bring more transparency to enable clinicians to understand and evaluate AI-based diagnostic systems.

1.1.7 Longitudinal analysis and progression monitoring

Longitudinal surveillance research in internet-based Parkinson's disease (PD) diagnosis is not common. Cross-sectional discrimination between healthy controls and PD patients was used in most previous research to probe the use of advanced-to-moderate disease states [22]. While best suited to differential diagnostics, they cannot fulfill the large clinical need for early diagnosis and ongoing monitoring of disease progression when the drug might be most beneficial.

Current systematic reviews have called attention to the necessity of devising strong longitudinal frameworks that could identify minimal changes in motor and behavioral changes over time [23]. These frameworks would enable earlier diagnosis, enable personalized treatment regimens, and ensure better long-term outcomes for the patient in the long run.

1.2 Research Gap

Comprehensive analysis of existing literature reveals several critical gaps that limit the clinical utility and real-world deployment of current digital PD assessment systems:

Gap 1: Limited Integration of Multiple Drawing Modalities

More recent efforts in digital evaluation of Parkinson's disease have been focused on single independent drawing tests, typically testing spiral drawings or handwriting samples in isolation [24]. Combined approaches have been tried in other research efforts, but the full cross-validation of multimodal drawing types, wave and spiral forms, within a unifying assessment paradigm has not been adequately addressed. This limitation hinders the full capture of the richness of several motor impairment factors and may lower the overall diagnostic stability of existing methods.

Gap 2: Insufficient Validation of Touchscreen-Based Platforms

Although digitizer-based technology has been widely tested for medical applications, commercial touchscreens pose unique technical problems not yet sufficiently investigated [25]. Heterogeneity of sampling rate, touch interface, and pressure sensitivity among manufacturers could significantly affect measurement reproducibility and compromise diagnostic reliability.

Gap 3: Inadequate Focus on Early-Stage Detection

Existing literature has mostly focused on classifying clinically diagnosed patients compared to healthy controls, typically in moderate-to-advanced stages with clear motor dominance [26]. This methodology is not, however, suited to meet the critical clinical need to identify early-stage disease in prodromal phases when earlier treatment would have the most likelihood of slowing disease deterioration.

Gap 4: Limited Real-World Deployment and Usability Studies

Most studies have been conducted in controlled laboratory settings with minimal investigation of actual deployment, usability, and performance in heterogeneous groups with differing degrees of technology literacy [27]. This is a significant barrier to the translation of research into clinical practice.

Gap 5: Lack of Comprehensive Longitudinal Frameworks

Most of the existing research has employed cross-sectional designs compared with healthy controls at one point *in* time. Very few studies have employed combined longitudinal designs that are crucial *to* monitor *disease* progression in individual assessment sessions, thereby serving useful inputs for early intervention and focused clinical management [28].

Research Gap Comparison Matrix

System/Research	Touchscreen Platform	Multi-modal Assessment	Early-stage Detection	Real-world Validation	Longitudinal Analysis
San Luciano et al. [12]	✗	✗	✗	✗	✗
Saleh et al. [13]	✗	✗	✗	✗	✗
Bernardo et al. [15]	✗	✗	✓	✗	✗
Burq et al. [19]	✗	✗	✗	✗	✗
Proposed System	✓	✓	✓	✓	✓

Figure 1 : Research Gap Comparison Matrix

1.3 Research Problem

Primary Research Problem

How would touchscreen consumer-level devices be utilized to develop an integrated, accessible instrument for evaluation that could detect early-stage Parkinson's disease visuospatial and motor dysfunctions through composite analysis of wave and spiral drawing patterns?

Specific Problem Dimensions:

Technical Feasibility Challenges:

- Validating consumer touchscreen devices for precise kinematic measurements comparable to clinical-grade digitizers.
- Developing robust signal processing algorithms to accommodate variations in device specifications and sampling rates.
- Implementing real-time processing capabilities suitable for mobile deployment.

Diagnostic Performance Requirements:

- Achieving diagnostic sensitivity and specificity appropriate for early-stage PD detection during prodromal phases.
- Designing optimal feature engineering approaches that combine kinematic, geometric, and temporal characteristics.
- Integrating multiple drawing modalities within a unified classification framework.

Clinical Implementation Barriers:

- Addressing usability requirements across diverse user populations with varying technological proficiency.
- Ensuring cost-effectiveness to support large-scale deployment in primary care settings.
- Developing explainable AI frameworks that provide clinical interpretability for healthcare professionals.

Longitudinal Monitoring Capabilities:

- Implementing progression-tracking algorithms capable of detecting subtle changes over time.
- Establishing personalized baselines and methodologies for detecting deviations.
- Integrating assessment outputs with existing clinical workflows and electronic health record systems.

2. RESEARCH OBJECTIVES

2.1. Main Objective

To develop and evaluate a multimodal, AI-powered system for early detection of prodromal PD and continuous monitoring via four complementary data streams: wearable sensor data (tremor and activity), drawing biomarkers (spiral and wave tasks), speech biomarkers (pauses, slurring, monotony), and digital cognitive tests (memory, attention, problem-solving, language). The modalities will each be processed individually and combined into a Multi-Modal Prediction Engine through ensemble and deep learning techniques. The system will utilize TinyML-capable light models for end-user device-based real-time, privacy-preserving analysis, augmented with explainable AI (XAI) methods such as attention maps and SHAP for clinician-readable, transparent decision support within a 12-month development cycle

2.2 Specific Objectives

To design and evaluate a comprehensive touchscreen-based assessment system for the detection of PD and monitoring of its progression through combined analysis of spiral and wave drawing patterns using off-the-shelf mobile devices over 12 months.

2.2.1 To design and implement a touchscreen-based data acquisition module (Months 1–3)

- Design and implement a data acquisition module that captures spiral and wave drawings at 60–120 Hz using both finger and stylus inputs.
- Securely store sessions and ensure participant stratification across PD severity levels; incorporate validated public datasets if primary data collection is limited.
- Each participant will complete multiple drawing sessions to establish reliable data patterns and baseline measurements.

2.2.2 To develop a robust feature engineering pipeline (Months 2–4)

- Develop a robust pipeline to extract 50–80 kinematic, geometric, temporal, and pressure-based features.
- Apply noise reduction and normalization techniques and validate measurement reliability using test-retest correlation coefficients above 0.75.

2.2.3 To build and optimize a Convolutional Neural Network (CNN)–based classification engine for handwriting and drawing analysis. (Months 4–8)

- Build and optimize a classification engine leveraging CNNs and Vision Transformers for disease detection and severity assessment.
- Target performance metrics: >80% accuracy, >75% sensitivity for early detection, and >85% specificity

2.2.4 To establish a longitudinal monitoring framework (Months 6–9)

- Establish a framework for tracking disease progression across multiple assessments using time-series analysis and change detection algorithms.
- Achieve statistically significant correlations ($p < 0.05$) with standard clinical measures.

2.2.5 To integrate explainable AI mechanisms (Months 8–10)

- Incorporate SHAP features importance and heatmap visualizations to generate clinician-friendly reports with confidence and uncertainty estimates.
- Achieve interpretability ratings above 4.0/5.0 from neurologists.

2.2.6 To deploy the integrated modules into a cross-platform mobile application (Months 9–11)

- Integrate modules into a user-friendly mobile application with intuitive drawing interfaces, secure data handling, and real-time feedback (<5 seconds per assessment).
- Ensure full compatibility across iOS and Android devices.

2.2.7 To evaluate the performance and usability of the system (Months 11–12)

- Conduct clinical validation with independent patient groups to ensure diagnostic accuracy against gold-standard assessments.
- Achieve a System Usability Scale (SUS) score above 70/100 and maintain inter-device correlation coefficients above 0.85.

3. METHODOLOGY

3.1 Methodology – Overall System Component

This research adopts a multimodal approach to designing an AI-powered diagnostic and monitoring system for Parkinson’s disease (PD). Instead of relying on a single source of information, the system combines complementary data streams to provide a more complete picture of both motor and cognitive symptoms. Four primary modalities are considered: wearable motion sensors, speech biomarkers, drawing tasks, and computerized cognitive assessments.

Wearable sensors, in the form of smartwatches or similar devices, continuously capture motion patterns such as tremor frequency, gait stability, and fine hand movements. These signals are preprocessed and transformed into clinically relevant features using lightweight TinyML algorithms that can operate efficiently on-device. Speech analysis contributes another perspective, where voice samples are examined for subtle abnormalities in pitch, tone, rhythm, and articulation. By applying advanced signal processing and natural language processing techniques, the system identifies speech-based biomarkers linked to early motor and cognitive decline.

The drawing component requires participants to complete standard spiral and wave tasks on touchscreen devices. These tasks provide a non-invasive way to measure motor control. Both image-based and temporal features—such as drawing velocity, curvature, entropy, and pressure variation—are extracted. Deep learning models, including convolutional networks and vision transformers, are then applied to detect atypical patterns that may indicate PD progression.

Digital cognitive tests further enrich the analysis by assessing higher-order functions like memory, attention, and executive control. Standardized tasks such as the Stroop Test and Trail Making Test capture response times, error rates, and switching ability, which serve as early indicators of cognitive impairment associated with PD.

Once extracted, features from all four modalities are transmitted securely to a centralized prediction engine. This engine employs ensemble machine learning and deep learning architectures to integrate multimodal data and compute individualized PD risk scores. To ensure clinical trust, explainable AI (XAI) techniques—including SHAP values and attention maps—highlight which features and modalities most strongly influence each prediction.

The complete system is deployed as a cross-platform mobile and web application, offering users real-time assessments, disease progression tracking, and visual reports. Designed with scalability and accessibility in mind, the platform supports both hospital-based use and remote monitoring in community or resource-limited settings. By unifying motor, speech, and cognitive signals into a single diagnostic pipeline, the methodology aims to provide accurate, interpretable, and practical tools for early detection and continuous monitoring of Parkinson’s disease.

3.2 Methodology – Drawing Biomarker Component

The drawing biomarker component focuses on using touchscreen-based spiral and wave tasks to detect early motor symptoms of Parkinson’s disease. Participants complete multiple drawing sessions on consumer-grade devices using either a finger or stylus, with data captured at high sampling rates. The collected drawings are processed to extract kinematic features (speed, acceleration, tremor), geometric features (spiral tightness, curvature), temporal features (rhythm, pauses, task duration), and spatial features (displacement, accuracy, pressure). These features are then fed into deep learning models—CNNs, Vision Transformers, and Bi-LSTMs—that work together in an ensemble to identify abnormal motor patterns. To ensure transparency, explainable AI methods such as SHAP and Grad-CAM highlight the most important features and drawing regions influencing predictions. The final system is deployed on a cross-platform mobile application with TensorFlow Lite, providing secure, real-time feedback and enabling both clinical use and home-based monitoring.

1. Hardware Setup and Configuration

- Use smartphones or tablets with touchscreen capability.
- Capture spiral and wave drawings using finger or stylus at 60–120 Hz.
- Store raw drawing data securely for later processing.

2. Data Collection Protocols

- Recruit participants from different PD severity stages and healthy controls.
- Tasks include spiral and wave drawings (multiple sessions per subject).
- Standardized conditions (lighting, seating, device placement).
- Use public datasets (HandPD, PaHaW) if recruitment numbers are low.
- Ensure anonymized and encrypted data storage.

3. Feature Extraction and Preprocessing

- Extract kinematic features: velocity, acceleration, jerk, tremor frequency.
- Extract geometric features: spiral tightness, curvature, trajectory entropy.
- Extract temporal features: rhythm, pauses, completion time.

- Extract spatial/pressure features: displacement, accuracy, pressure dynamics.
- Apply noise reduction, normalization, and test-retest reliability checks.
- Correlate features with clinical scores such as UPDRS-III.

4. Model Training and Optimization

- Build an ensemble deep learning model:
- CNN for trajectory sequences.
- Vision Transformer for spatial patterns.
- Bi-LSTM with attention for temporal dependencies.
- Train uses stratified cross-validation to avoid data leakage.
- Optimize with Adam optimizer and cosine annealing.
- Combine outputs through a weighted ensemble with uncertainty estimation.

5. Explainable AI (XAI) Integration

- Use SHAP to measure feature importance.
- Apply Grad-CAM to highlight critical drawing regions.
- Generate attention maps to show temporal and spatial focus.
- Produce clinician-friendly reports with confidence intervals and uncertainty estimates.

6. Mobile Deployment

- Deploy the module as a cross-platform app using React Native.
- Optimize models with TensorFlow Lite for on-device inference.
- Provide real-time feedback (<5 seconds per test).
- Use HIPAA-compliant AES-256 encryption for security.
- Enable use in both clinical and home environments.

7. Expected Outcomes

- Early detection of PD from subtle motor impairments.
- Longitudinal monitoring of disease progression.
- Clinically interpretable results supporting neurologists in decision-making

3.3 System Diagram

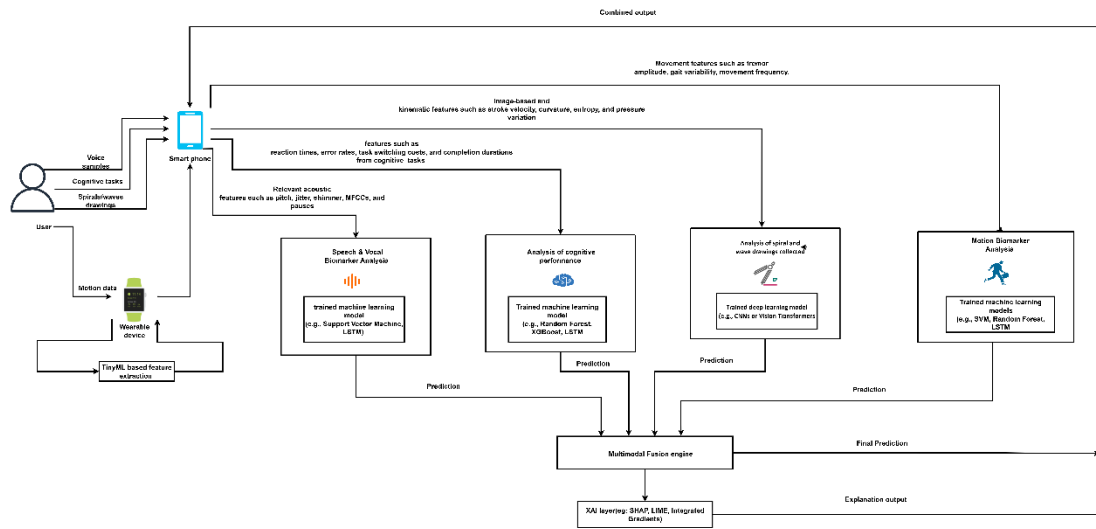


Figure 2 : System Architecture Diagram

3.4 System Component Diagram

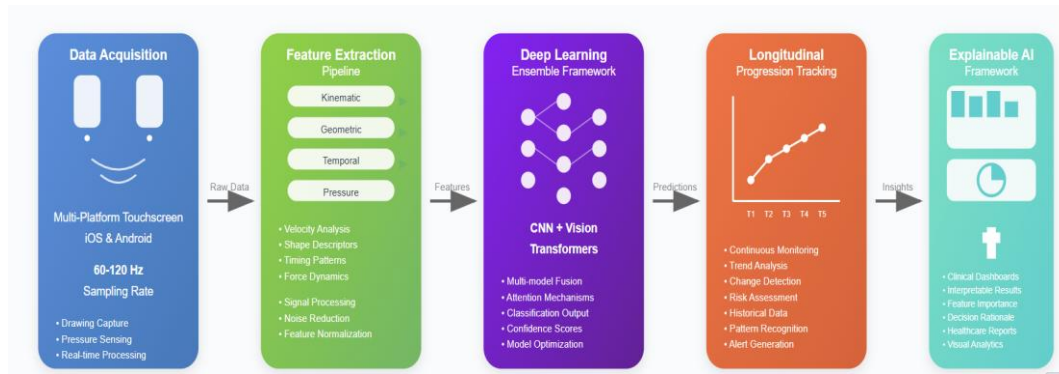


Figure 3 : Digital Drawing Analysis Pipeline

Drawing Analysis Component for Parkinson's Disease Detection

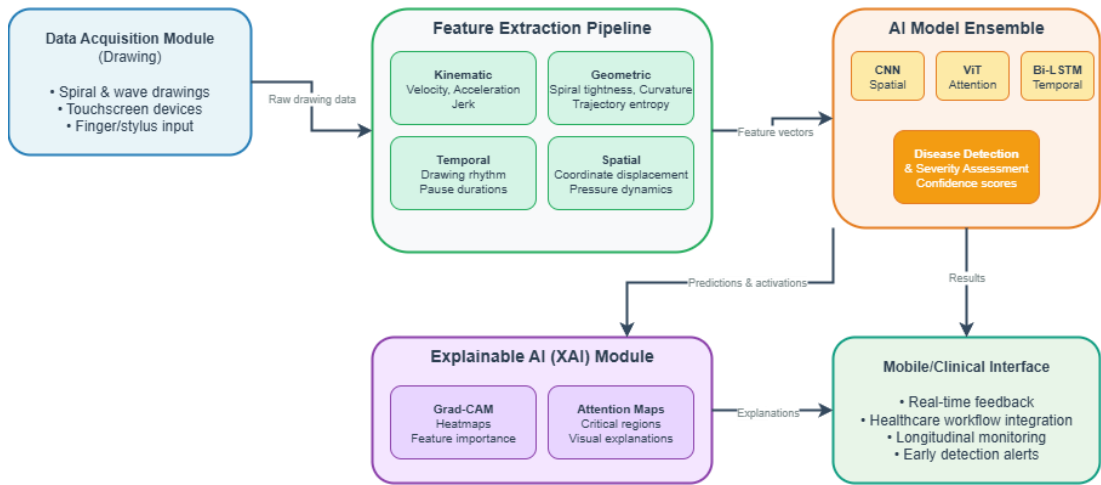


Figure 4 : System Component Diagram

3.5 Tools and Technologies

The project employs a stack of open-source and industry-standard tools for robustness and cost-effectiveness:

Tool/Technology	Description
Python	Versatile scripting language with libraries for data analysis.
Git	Distributed version control system.
VS Code	Lightweight, extensible code editor.
TensorFlow Lite	Lightweight ML framework for mobile/embedded devices.
Figma	Collaborative interface design tool.
Firebase	Backend-as-a-Service platform.
Scikit-learn	ML library for classical algorithms.
React Native	JavaScript framework for mobile application development.

Table 1 Tools and Technologies

4. COMMERCIALIZATION

Free Tier

The free option enables access to basic assessments and feedback via selected modules of the application. It is ideal for individuals looking for an initial screening or tracking tool for personal use. Limitations include less access to advanced analytics, multi-modal integration, and real-time AI-based prediction. Wearable device incorporation and explainable AI (XAI) outputs are disabled in this option, keeping the app light and accessible while providing basic early detection functionality.

Paid Tier (LKR 1990)

The paid tier frees the full potential of the app, e.g., complete multi-modal interaction. Wearable device integration is included in this level, with continuous, real-time monitoring of motor and cognitive signals along with it. This places the app far more proactively, beyond episodic self-identification to continuous digital biomarker acquisition. In addition, premium users have access to insights powered by AI, early detection forecasts, and explainable AI (XAI) results highlighting the ways different biomarkers are used to make the forecasts. This tier is designed to provide in-depth, clinically relevant tools to patients, caregivers, and clinicians.

This tiered model provides users with flexibility according to their needs, encourages sustainable app development, and facilitates continuous improvement. Updates will be released regularly, adding new features and enhancements, with premium levels offering extended monitoring, interpretability, and integration.

5. PROJECT REQUIREMENTS

5.1 Functional Requirements

Drawing Data Collection

- REQ-F001: The system shall capture spiral and wave drawings at a sampling rate of 60–120 Hz.
- REQ-F002: The system shall record velocity, acceleration, pressure, and temporal data during drawing tasks.
- REQ-F003: The system shall support both finger and stylus input on touchscreen devices.
- REQ-F004: The system shall automatically detect drawing completion and interruptions. In case of interruption, the system shall prompt the user to resume or restart.
- REQ-F005: The system shall store raw drawing data in a standardized format (JSON/CSV) to ensure reproducibility and interoperability.

Feature Processing

- REQ-F006: The system shall extract 50–80 features (kinematic, geometric, temporal, and spatial).
- REQ-F007: The system shall apply noise reduction using Gaussian filtering and signal processing techniques.
- REQ-F008: The system shall detect tremor frequencies in the 3–8 Hz range using spectral analysis.
- REQ-F009: The system shall normalize extracted features (e.g., z-score normalization) for consistency across users.

Classification System

- REQ-F010: The system employ ensemble models (CNN, Vision Transformer, LSTM) for PD detection.
- REQ-F011: The system shall output probability scores with 95% confidence intervals for each prediction.
- REQ-F012: The system shall achieve >85% accuracy, >80% sensitivity, and >90% specificity on validation datasets.
- REQ-F013: The system shall provide explainability features, highlighting influential features or drawing segments contributing to classification.

Clinical Reporting

- REQ-F014: The system shall generate visual overlays showing abnormal drawing regions.
- REQ-F016: The system shall allow longitudinal tracking of disease progression across multiple sessions.
- REQ-F017: The system shall export reports in PDF and HL7-compliant XML for integration into hospital systems.

5.2 Non-Functional Requirements

Performance

- REQ-NF001: Drawing analysis shall complete within 30 seconds.
- REQ-NF002: The system shall support at least 100 concurrent users, with scalability to 500 users.
- REQ-NF003: The UI shall respond within 2 seconds.

Reliability

- REQ-NF004: The system shall maintain 99.5% uptime during operational hours.
- REQ-NF005: The system shall perform automatic data backups every 15 minutes.
- REQ-NF006: The system shall recover from failures within 5 minutes, with disaster recovery procedures defined.

Security

- REQ-NF007: All data shall be encrypted using AES-256 in transit and at rest.
- REQ-NF008: Providers shall log in using multi-factor authentication.
- REQ-NF009: The system shall comply with HIPAA and GDPR regulations.
- REQ-NF010: The system shall implement role-based access control (RBAC) for patients, providers, and administrators.

Usability

- REQ-NF011: The system shall achieve a System Usability Scale (SUS) score >75.
- REQ-NF012: Elderly users shall complete tasks with >90% success rate and <5% error rate.
- REQ-NF013: The system shall support accessibility features (large buttons, audio guidance, screen readers).
- REQ-NF014: The system shall support multi-language interfaces for patient accessibility.

Maintainability & Scalability

- REQ-NF015: The system shall allow updates and patches without downtime (rolling updates via Kubernetes).
- REQ-NF016: The backend shall scale horizontally to support increasing patient numbers.

5.3 User Requirements

Patients

- Easy-to-use interface suitable for elderly and impaired users.
- Clear visual, audio, or voice-assisted instructions.
- Progress tracking with simple charts and reminders for regular use.
- Multilingual support (English, Sinhala, Tamil).

Healthcare Providers

- Access to detailed diagnostic reports with clear explanations.
- Integration with existing hospital/clinical systems.
- Tools to track patient progress over time.

Administrators

- User and access management features.
- System usage analytics and monitoring.
- Strong privacy and security compliance.

5.4 System Requirements

Hardware

- Smartphone or tablet with sufficient memory and storage.
- Touchscreen support with good display quality.
- Microphone for voice input.
- Internet connectivity for updates and data sync.

Software

- Support for Android/iOS platforms.
- Machine learning frameworks for model deployment.
- Secure local and cloud database for data storage.
- Security & Performance
- Encrypted and backed-up data storage.

5.5 Wire Frame

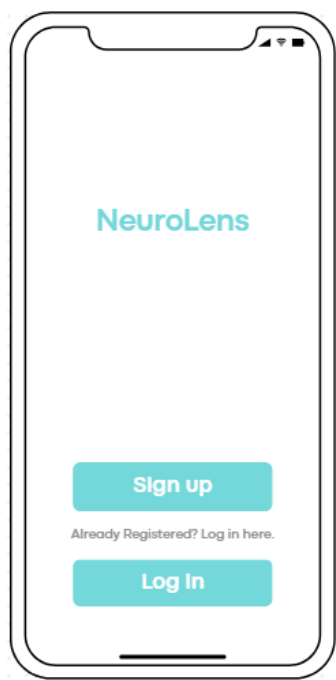


Figure 5 : On Boarding Screen

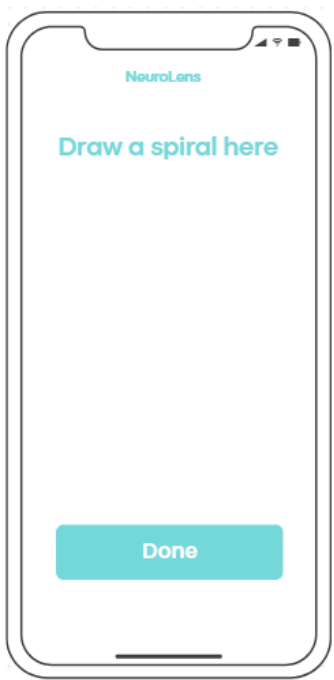


Figure 6 : Canvas Screen

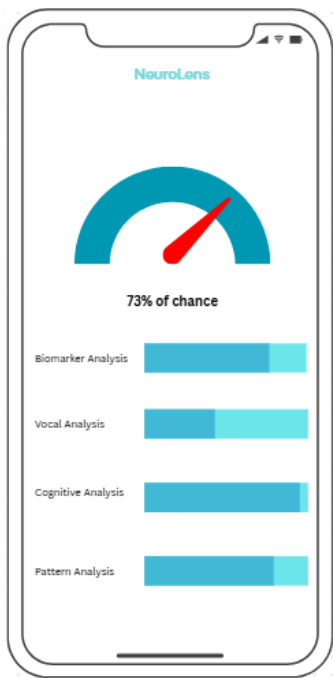


Figure 7:Final Output Screen

5.6 Gantt Chart



Figure 8 : Gantt Chart

5.7 Work Breakdown Structure (WBS)

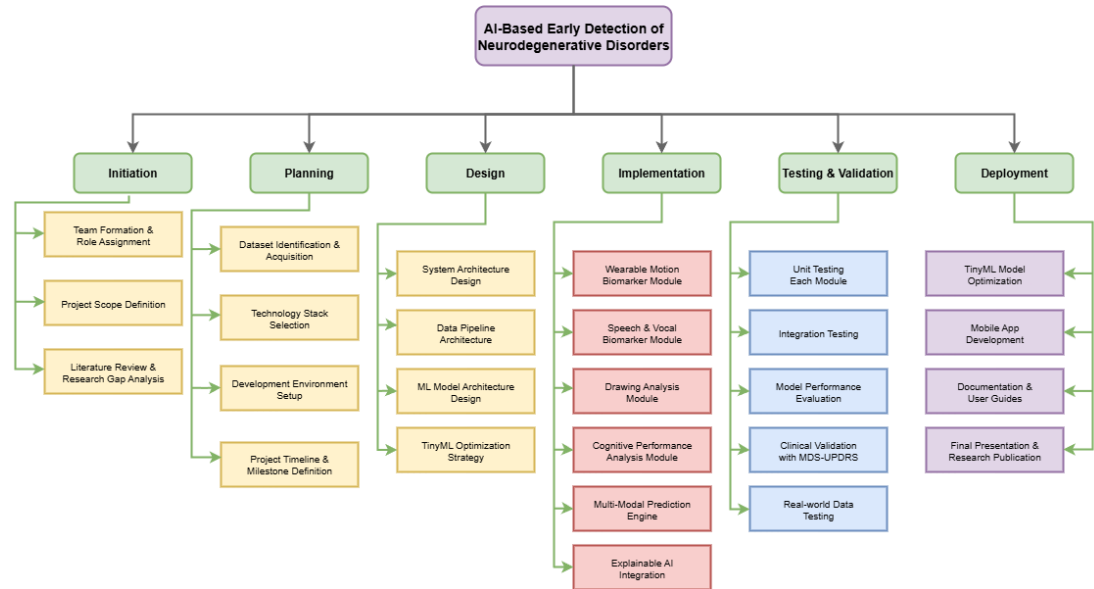


Figure 9 : Work Break Down Structure(WBS)

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